Walks, Paths, and Connectivity Read from an Adjacency Matrix

A Study Note

Abstract

A finite simple undirected graph with N vertices can be described by its adjacency matrix A, a square table of zeros and ones. This note answers three questions about it. First, the entry in row i and column j of the k-th power A^k counts the walks of length k from vertex n_i to vertex n_j. It counts walks, not paths, and the note shows by a counterexample why the difference matters. Second, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive. Both statements are proved in full. Third, a path between n_i and n_j is a walk from n_i to n_j in which no vertex occurs twice. The note follows the numbering and build order of the course textbook, sets every new idea in a single story about a village, and works two small examples with every matrix written out and computed by script.

1 Introduction

1.1 The three questions and where they came from

A graph is a set of points, called vertices, together with some lines, called edges, each joining two vertices. When a graph has N vertices, one way to describe it completely is a square table of N rows and N columns, the *adjacency matrix* A, which has a one in row i and column j when vertices number i and number j are joined by an edge and a zero otherwise. This note asks what such a table can tell us, and answers three questions.

Q1 What does the entry $(A^k)_{ij}$ of the k-th power of A represent?

Q2 How can one determine from A alone whether the graph is connected?

Q3 What is the definition of a path between vertices n_i and n_j?

Questions Q1 and Q2 are written on a lecture whiteboard that was photographed twice [4, 5]. On the board, Q1 reads "What does $A^{\cdot}_{ij}$ represent" with a small mark in the place of the exponent, and Q2 reads "How to know if graph is connected or not". The board also gives the definition "Graph is called connected if for any n_i, n_j there exists a path between nodes", a drawing of a graph on three vertices, and a three by three matrix labelled "adj." [4, 5]. Question Q3 was added by the assignment text, which we know only as it was reported to us and do not reproduce here. Appendix B transcribes the board line by line.

1.2 One story for the whole note

Every definition, theorem, test, and worked example in this note is preceded by a paragraph headed "In plain words". Those paragraphs are told in one story and use no symbols. The story is a small village. The village has houses. Roads join the houses in pairs. Each road joins two different houses. Two houses have at most one road between them. Each road can be driven in both directions. Table 1 fixes the story words once; each story word stands for exactly one technical term, and the technical term is defined at the place named in the last column.

Table 1: The story dictionary. A story word is used only in the plain words paragraphs, and a technical word is never used there.

Story word	Technical term	Defined in
village	graph	Definition 2.1
house	vertex	Definition 2.1
road	edge	Definition 2.1
neighbours	adjacent vertices	Definition 2.5
table of roads	adjacency matrix	Definition 2.9
number of roads at a house	degree	Definition 2.14
table of totals	the sum S	Definition 4.1
trip	walk	Definition 2.19
number of roads driven	length of a walk	Definition 2.19
round trip	closed walk	Definition 2.23
trip with no road twice	trail	Definition 2.27
route	path	Definition 2.31
round trip with no road twice	cycle	Definition 2.36
ring	simple cycle	Definition 2.36
shortest trip between two houses	distance	Definition 2.40
village in one piece	connected graph	Definition 2.48
piece of a village	component	Definition 2.52

A trip may pass the same house twice and may drive the same road twice. A route never passes a house twice. This difference is the main source of mistakes about the matrix, and the whole note keeps the two words apart.

1.3 Conventions and assumptions

The photographs and the assignment text leave some points open. We settle each point as follows and refer to these items by number.

1. *The graphs.* The graph is finite, simple (no edge from a vertex to itself and no two edges with the same two ends), and undirected. The assignment text says so as reported to us, and it agrees with the matrix on the board, which is symmetric with zeros on the diagonal [4].

2. *The exponent.* The mark above A in question Q1 is read as a positive integer exponent k, because the assignment text asks about A^k. In both photographs the mark looks like a small n or a pair of quotation marks, not clearly a k [4, 5]. The answer given below holds for every positive integer exponent, so it does not depend on which letter was written (Section 7.2).

3. *The example graph.* On the board the drawing and the matrix disagree. The drawing shows three lines: between vertices 1 and 2 (with an arrowhead at 2), between 1 and 3, and between 2 and 3. The matrix has rows 0 1 0, 1 0 1, 0 1 0, which records only the edges between 1 and 2 and between 2 and 3 [4, 5]. The board states that the graph is given through its adjacency matrix [4], so the matrix defines the example graph, and the arrowhead is treated as a stray mark because the matrix is symmetric. We report the mismatch and make no claim about which of the two the lecturer intended (Section 7.1).

4. *The word "path".* In the course, a path has no repeated vertex, as reported to us. The textbook calls such a walk a *simple path* [9, Definition 1.2.13, p. 22], and this note uses

"path" for it throughout. The board does not say which meaning of "path" it uses. The textbook's own word "path" means something weaker, which this note calls a *trail*; Section 5 compares the three usages.

5. *Notation.* The vertices are $n_1, \ldots, n_N$ as on the board, where N is the number of vertices; the textbook writes v_i and n. The board writes (G, V) for the graph and $e_{ij} \in V$ in the line that defines A_{ij}; its V there stands for the set of edges, which the textbook writes E in $G = (V, E)$ [4]. This note writes $G = (V, E)$, calls an edge $n_i n_j$ (or e_{ij}), and writes $|E|$ for the number of edges. The letters i, j, r are vertex numbers, k is an exponent or the length of a walk, I is the identity matrix, and S is the sum defined in Definition 4.1.

6. *Length zero.* A single vertex counts as a walk of length 0. The textbook does not say whether this is allowed; we allow it because the textbook's own distance property $d(u, u) = 0$ needs it [9, printed p. 23]. A walk of length 0 is not counted as a cycle.

7. *Page numbers.* Pages of the textbook are printed page numbers; the PDF page is one higher.

1.4 What the textbook contains and what it does not

The course textbook supplies the definitions and their numbers [9]. A full-text search of the whole book for the adjacency matrix finds it only on printed pages 7, 11, 12, 34, and 61, and finds neither a theorem that powers of the adjacency matrix count walks nor a matrix test for connectivity [9]. We did not read every page, so this is a statement about the search and not a proof that the book has no such result. Both results are therefore taken from other sources and proved in full here: the walk-counting theorem from [2] and the connectivity test, whose statement is in [7] and [1]. Where this note proves a statement itself, it says so in the first sentence of the result and names a source that also states it.

1.5 The answers in brief

Q1 For $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j. It counts walks, not paths (Theorem 3.5 and proposition 3.10).

Q2 For $N \geq 2$, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive; equivalently, if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 4.6).

Q3 A path between n_i and n_j is a walk from n_i to n_j whose vertices are pairwise distinct (Definition 2.31); Section 5 compares this with the other usages.

1.6 Organisation

Section 2 builds the definitions in the textbook's order, with its numbers. Section 3 answers Q1 and Section 4 answers Q2, including a breadth-first search alternative. Section 5 answers Q3 by comparing conventions. Section 6 works two examples on three vertices, one connected and one not. Section 7 states the limits, and Section 8 gives the direct answers. The appendices hold the board transcription, a table of textbook numbers, the longer derivations, and the sources and scripts. Directed graphs, weighted graphs, and eigenvalue methods are outside the scope of this note.

2 Background: graphs, walks, and connectedness

This section builds the definitions in the order of the course textbook [9], with the textbook's numbers beside them. Each definition is followed by a remark on what it allows and forbids, an example that satisfies it, and an example that does not.

2.1 Notation

Let $G = (V, E)$ be a graph with vertex set $V = \{n_1, \ldots, n_N\}$ and edge set E; let u, v, w denote vertices and e, e' denote edges. The *board graph* B is the graph with $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2,\ n_2n_3\}$, the graph whose adjacency matrix is written on the board (Item 3). A walk in B is written by the numbers of its vertices, so 1-2-3 means n_1, n_2, n_3.

2.2 Graphs, adjacency, and the adjacency matrix

In plain words. A village is a list of houses and a list of roads. Every road joins two different houses. No two roads join the same two houses. A road has no direction.

Definition 2.1 (graph; textbook Definition 1.1.1, p. 9)**.** A *simple undirected graph* is a pair $G = (V, E)$ where V is a finite nonempty set, whose elements are the *vertices*, and E is a set of unordered pairs of distinct elements of V, whose elements are the *edges* [9, p. 9][2, Definition 1.1.1].

Remark 2.2. The definition forbids an edge from a vertex to itself (the two elements of the pair must be distinct) and a repeated edge (E is a set). It forbids direction, because the pair is unordered, so n_in_j and n_jn_i are one edge. It allows a vertex with no edge at all.

Example 2.3. $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2,\ n_2n_3\}$: the board graph B, three houses in a row.

Non-example 2.4. Two vertices joined by two edges, or a vertex joined to itself. The first repeats an edge and the second is a loop, so neither is a simple graph.

In plain words. Two houses are neighbours when one road joins them directly. Nothing is said about longer trips. A house is never its own neighbour.

Definition 2.5 (adjacent; textbook Definition 1.1.4, p. 11)**.** Vertices u and v are *adjacent* if $uv \in E$; each is then a *neighbour* of the other [9, p. 11].

Remark 2.6. Adjacency is symmetric, and no vertex is adjacent to itself because there are no loops. It says nothing about connections through other vertices.

Example 2.7. In B the vertices n_1 and n_2 are adjacent.

Non-example 2.8. In B the vertices n_1 and n_3 are not adjacent, although a trip along two edges joins them.

In plain words. Make a square table with one row and one column for every house. Write a one where the row house and the column house share a road. Write a zero everywhere else. This table describes the whole village.

Definition 2.9 (adjacency matrix; textbook p. 11, after Definition 1.1.6)**.** Let G be a graph with vertices $n_1, \ldots, n_N$. Its *adjacency matrix* is the $N \times N$ matrix $A = (A_{ij})$ with $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise. The textbook gives this rule without a definition number [9, p. 11]; the board states it as $A_{ij} = 1$ if $e_{ij} \in V$ and 0 if $e_{ij} \notin V$, with V standing for the edge set (Item 5) [4, 5].

Remark 2.10. The only entries allowed are 0 and 1. The matrix depends on how the vertices are numbered: renumbering moves rows and columns together.

Example 2.11. The board matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ is the adjacency matrix of B: $A_{12} = A_{21} = 1$, $A_{23} = A_{32} = 1$, and every other entry is 0.

Non-example 2.12. The matrix $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ is not the adjacency matrix of any simple undirected graph, because $A_{12} = 1$ but $A_{21} = 0$. A matrix with a 1 on its diagonal (a loop) or an entry 2 (a double edge) is not one either.

In plain words. The table of roads has easy features. It looks the same when flipped over the line of squares from top left to bottom right. That line holds only zeros. Any such table of zeros and ones comes from some village.

Proposition 2.13 (properties of A)**.** *The adjacency matrix A of a graph is symmetric ($A_{ij} = A_{ji}$ for all i, j) and has zero diagonal ($A_{ii} = 0$ for all i). Conversely, every $N \times N$ matrix of zeros and ones that is symmetric with zero diagonal is the adjacency matrix of some graph with numbered vertices.*

Proof. The textbook states both parts [9, pp. 11–12]; see also [2, Section 1.1] and [7, property 9]. Adjacency is symmetric and no vertex is adjacent to itself (Remark after Definition 2.5), which gives the first part. For the converse, join n_i and n_j exactly when the entry is 1. Symmetry makes this a set of unordered pairs, and the zero diagonal excludes loops. □

In plain words. The number of roads at a house tells how busy that house is.

Definition 2.14 (degree; textbook Definition 1.2.1, p. 16)**.** The *degree* of a vertex n_i is the number of edges incident to n_i. The textbook writes it $d_G(v)$ or $d(v)$ [9, p. 16]; this note writes $\deg(n_i)$ so that the letter d is kept for distance (Definition 2.40).

Remark 2.15. In a simple graph the degree is also the number of neighbours, because each neighbour is joined to n_i by exactly one edge.

Example 2.16. In B: $\deg(n_1) = 1$, $\deg(n_2) = 2$, $\deg(n_3) = 1$.

Non-example 2.17. $\deg(n_1) \neq 2$ in B, although n_2 and n_3 can both be reached from n_1: only edges at n_1 count, and n_3 is reached through n_2.

In plain words. Add up one row of the table. You get the number of roads at that house. Add up the whole table. Every road has been counted twice, once from each end.

Proposition 2.18 (row sums are degrees)**.** *Row i of A, and column i, sums to $\deg(n_i)$, and the sum of all entries of A is $2|E|$.*

Proof. This is proved here; [7, property 8] states the row sum for graphs that may also have loops, and the textbook's Theorem 1.2.1 (Euler) states that the degrees sum to $2|E|$ [9, p. 16]. Row i has a 1 in column j exactly when n_j is a neighbour of n_i. In a simple graph each neighbour is joined to n_i by exactly one edge and no edge joins n_i to itself, so the number of ones in row i is the number of edges at n_i, which is $\deg(n_i)$. Column i is the same by symmetry (Proposition 2.13). Adding the row sums over all i counts every edge twice, once from each end, so the total is $2|E|$. □

For B the row sums are $1, 2, 1$ and the total is $4 = 2|E|$ with $|E| = 2$.

2.3 Walks, trails, paths, and cycles

In plain words. A trip is a list of houses, one after another. The next house is always at the other end of a road from the house before. You may pass the same house twice. You may drive the same road twice. The length of a trip is the number of roads you drove. A road driven twice counts twice. Staying at home is also a trip, of length zero.

Definition 2.19 (walk; textbook Definition 1.2.10, p. 21)**.** A *walk of length k from u_0 to u_k* is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_i = u_{i-1}u_i$ for $1 \leq i \leq k$. It is abbreviated by its vertex sequence $u_0, u_1, \ldots, u_k$. The *length* of the walk is the number k of its edges, each edge counted as many times as it occurs [9, pp. 21–22][3, Definition 2.9.1].

Remark 2.20. Vertices and edges may repeat in a walk [9, p. 22]. The definition forbids only one thing: two consecutive vertices that are not joined by an edge. By Item 6 the case $k = 0$, a single vertex, is a walk. In a simple graph an edge is fixed by its two ends, so a walk is determined by its vertex sequence; hence a walk of length k from n_i to n_j is the same as a sequence $w_0, \ldots, w_k$ of vertices with $w_0 = n_i$, $w_k = n_j$ and w_{t-1} adjacent to w_t for every t. This is a consequence of Definition 2.1 and Definition 2.19 [3, Definition 2.9.1(a)], and the proof of Theorem 3.5 counts such sequences.

Example 2.21. In B the sequence 1-2-1-2-3 is a walk of length 4 from n_1 to n_3: its steps are the edges n_1n_2, n_2n_1, n_1n_2, n_2n_3. The vertices n_1 and n_2 occur twice, and the edge n_1n_2 is used twice. The textbook's example $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ in its Fig. 1.2.5 is a walk, of length 8 [9, p. 22].

Non-example 2.22. In B the sequence 1-3 is not a walk, because n_1n_3 is not an edge of B ($A_{13} = 0$). The textbook's $v_3, v_1, v_4, v_2, v_3, v_5$ is not a walk of its graph [9, p. 22].

In plain words. A round trip ends at the house where it began. That is the only new rule.

Definition 2.23 (closed walk; textbook Definition 1.2.11, p. 22)**.** A walk $u_0, \ldots, u_k$ is *closed* if $u_0 = u_k$ [9, p. 22].

Remark 2.24. A closed walk may repeat anything a walk may repeat. A walk of length 0 is closed. In a simple graph no closed walk has length 1 (it would be a loop), and a closed walk of length 2 is u, w, u, which uses one edge twice.

Example 2.25. In B the walk 2-1-2 is closed, of length 2.

Non-example 2.26. In B the walk 1-2-3 is not closed: it starts at n_1 and ends at n_3.

In plain words. Now add one rule. Never drive the same road twice. You may still pass a house twice, if you come back by a different road. Books use different words for such a trip. Section 5 lists them.

Definition 2.27 (trail; textbook Definition 1.2.12, p. 22, where it is called "path")**.** A walk $u_0, \ldots, u_k$ is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct [9, p. 22]. English usage also calls this a trail [3, Definition 3.4.5][8].

Remark 2.28. A trail may repeat vertices, but not edges. A vertex can recur in a trail only if the trip left it by one edge and returned by another. When the textbook writes "path" for this object, this note writes "trail" or "the textbook's path".

Example 2.29. The textbook's $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail: its edges are distinct, but v_3 occurs twice [9, p. 22].

Non-example 2.30. The textbook's $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ is a walk but not a trail [9, p. 22]. In B the walk 1-2-1-2-3 is not a trail, because the edge n_1n_2 occurs twice.

In plain words. Now add a stronger rule. Never pass the same house twice. Then you also never drive a road twice. Such a trip goes straight from start to end with no loop. It is called a route.

Definition 2.31 (path, and path between n_i and n_j; textbook Definition 1.2.13, p. 22, where it is called "simple path")**.** A walk $u_0, \ldots, u_k$ is a *path* if its vertices $u_0, \ldots, u_k$ are pairwise distinct [9, p. 22][3, Definition 2.9.1(c)]. A *path between n_i and n_j* is a path with $u_0 = n_i$ and $u_k = n_j$; its length is k.

Remark 2.32. This is the answer to question Q3, under Item 4. A path forbids repeated vertices, and therefore repeated edges (Proposition 2.35). It allows length 0 (a single vertex) and length 1 (one edge). It does not allow a closed walk of positive length, because its first and last vertices would be equal; so for $i = j$ the only path between n_i and n_i is the single vertex n_i.

Example 2.33. In B the sequence 1-2-3 is a path between n_1 and n_3, of length 2, and it is the only one. The textbook's v_1, v_3, v_4, v_5 is a simple path [9, p. 22].

Non-example 2.34. In B the walk 1-2-3-2 is not a path, because n_2 occurs twice, and neither is 1-2-1-2-3 although it joins n_1 to n_3. The textbook's $v_1, v_3, v_2, v_4, v_3, v_5$ is not a simple path [9, p. 22].

In plain words. A route never drives a road twice. Here is why. A road driven twice would bring you to a house for the second time.

Proposition 2.35 (every path is a trail)**.** *In a simple graph, a walk whose vertices are pairwise distinct has pairwise distinct edges.*

Proof. This is proved here; [3, Exercise 2.9] states it as an exercise. Let $u_0, \ldots, u_k$ have distinct vertices and take two steps $a < b$, the edges $u_{a-1}u_a$ and $u_{b-1}u_b$. The vertex u_b has an index larger than a, so it differs from u_a and from u_{a-1}. Thus u_b is an end of the second edge and not an end of the first, and the two edges are different. □

In plain words. A ring is a round trip that passes no house twice. Only the start may appear twice, because it is also the end. The textbook also names a looser round trip. It may pass a house twice, but it never drives a road twice.

Definition 2.36 (cycle and simple cycle; textbook Definitions 1.2.14 and 1.2.15, p. 22)**.** A trail that is a closed walk is called a *cycle* (the textbook also says *closed path*) [9, Definition 1.2.14]. A cycle is *simple* if only its first and last vertices repeat; the simple cycle on n vertices is written C_n, $n \geq 3$ [9, Definition 1.2.15, p. 22].

Remark 2.37. A cycle in the textbook sense may repeat vertices (a figure eight is one); it forbids repeated edges and open walks. A simple cycle forbids every repeated vertex except the closing one. A closed trail of length 0 is not counted as a cycle (Item 6). English usage calls only the simple cycle a "cycle" [3, Definition 2.10.1(b)], so Section 5 keeps the two senses apart. Cycles are not needed for the main results; they are defined because the textbook's build order includes them.

Example 2.38. The textbook's v_1, v_3, v_4, v_2, v_1 is a simple cycle, and $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ is a cycle in the sense of Definition 2.36 but not a simple cycle, because v_3 occurs three times [9, p. 22]. In the triangle drawn on the board (the graph T with edges n_1n_2, n_1n_3, n_2n_3) the walk 1-2-3-1 is a simple cycle of length 3.

Non-example 2.39. In B the closed walk 1-2-1 is not a cycle, because the edge n_1n_2 occurs twice. In fact B has no cycle: a trail in B has at most $|E| = 2$ edges, a closed walk of length 1 does not exist, and a closed walk of length 2 repeats its edge (Remark after Definition 2.23); a computer enumeration of the closed walks of length up to 6 agrees (Appendix D).

In plain words. Take two houses and look at the shortest trip between them. Count the roads on it. That number tells how far apart the houses are. Staying at home is a trip of length zero, so a house is zero far from itself. If no trip exists, the number is declared to be infinite.

Definition 2.40 (distance; textbook Definition 1.2.16, pp. 22–23)**.** The *distance* $d(u, v)$ between vertices u and v is the length of a shortest (u, v)-path in the textbook's sense (a trail) if at least one exists, and $+\infty$ otherwise [9, pp. 22–23]. By Proposition 2.47 the value is the same whether trails, paths, or walks are used. The textbook lists three properties [9, p. 23]:

1. $d(u, v) \geq 0$, and $d(u, v) = 0$ if and only if $u = v$;
2. $d(u, v) = d(v, u)$;
3. $d(u, v) \leq d(u, w) + d(w, v)$.

Hence, when any two vertices are joined by a path, the pair (V, d) is a metric space [9, p. 23]. Proposition C.2 derives the three properties from the definition.

Remark 2.41. The value $+\infty$ is not an integer; it is how a missing connection shows up. No value is negative, and no value is anything other than the length of an actual trip.

Example 2.42. In B: $d(n_1, n_2) = 1$, $d(n_2, n_3) = 1$, $d(n_1, n_3) = 2$, and $d(n_i, n_i) = 0$.

Non-example 2.43. In B the number 4 is not $d(n_1, n_3)$, although a walk of length 4 joins them: distance is the minimum, not any length. In the graph with three vertices and the single edge n_1n_2 (Example 6.3) the distance $d(n_1, n_3)$ is not a number: it is $+\infty$.

2.4 From a walk to a path

In plain words. Suppose you can drive from one house to another. Then you can do it without passing any house twice. Look at your trip. If it passes some house twice, you went round a loop. Cut the loop out. What is left is still a trip between the same two houses, and it is shorter. Keep cutting until no loop is left.

Lemma 2.44 (every walk contains a path)**.** *Let u and v be vertices, equal or not. If there is a walk of length k from u to v, then there is a path from u to v of length at most k, whose vertices and edges all belong to the walk.*

The textbook proves the case $u \neq v$ by induction on the length of the walk, without stating the bound on the length [9, Lemma 1.2.1, p. 23]; the general statement with the bound is [3, Corollary 2.9.7, p. 39]. The proof below follows the textbook's method.

Proof. We prove by strong induction on k the statement $S(k)$: every walk of length k, between any two vertices u and v, contains a path between u and v of length at most k.

Base case $k = 0$. A walk of length 0 is a single vertex, so it has no repeated vertex and is itself a path, of length $0 \leq 0$.

Induction hypothesis. Let $k \geq 1$ and assume $S(m)$ for every $m < k$.

Step. Let $u_0, \ldots, u_k$ be a walk with $u_0 = u$ and $u_k = v$. If its vertices are pairwise distinct, it is a path of length k, and we are done. Otherwise there are indices $i < j$ with $u_i = u_j$. Form $W' = u_0, \ldots, u_i, u_{j+1}, \ldots, u_k$ (if $j = k$ the part after u_i is empty). Then W' starts at u. It ends

at v: if $j < k$ its last vertex is u_k, and if $j = k$ it is $u_i = u_j = u_k$. Every two consecutive vertices of W' are consecutive in the old walk, except the pair $(u_i, u_{j+1}) = (u_j, u_{j+1})$, which is also a step of the old walk. So W' is a walk from u to v, of length $i + (k - j) < k$. By $S(m)$ with $m = i + (k - j) < k$, W' contains a path from u to v of length at most $m < k$, and every vertex and edge of W' belongs to the old walk. This proves $S(k)$, and by induction the lemma. □

Example 2.45. In B the walk 1-2-1-2-3 has $u_0 = u_2 = n_1$; cutting gives the path 1-2-3. The closed walk 2-1-2 is cut down to the single vertex n_2, a path of length 0. For the textbook's walk of length 8 two cuts give walks of length 5 and then 2, the path v_1, v_3, v_5 (Appendix D).

In plain words. A route passes each house at most once. So a route cannot be longer than the number of houses minus one. Hence if any trip exists, a short trip exists.

Corollary 2.46 (bound on path length)**.** *In a graph with N vertices every path has at most $N - 1$ edges. Hence if some walk joins u and v, a walk of length at most $N - 1$ joins them.*

Proof. A path of length k lists $k + 1$ pairwise distinct vertices, and there are only N vertices, so $k + 1 \leq N$ [3, Section 2.9.4]. For the second part, apply Lemma 2.44: the path inside the walk has length at most that of the walk and at most $N - 1$, and a path is a walk. □

In B we have $N = 3$ and the longest path has two edges.

Proposition 2.47 (shortest walk, trail, and path agree)**.** *For vertices u, v the lengths of a shortest (u, v)-walk, of a shortest (u, v)-trail, and of a shortest (u, v)-path are equal, and all three are infinite together.*

This is proved here (Appendix C.5); the remark that a shortest walk is a shortest path is also made in [3, Section 2.9.4].

2.5 Connected graphs and components

In plain words. Look at a village. Sometimes you can reach every house from every other house. Then the village is in one piece. If one house is cut off, it is not in one piece.

Definition 2.48 (connected; textbook Definition 2.1.1, p. 48; board)**.** A graph G is *connected* if for any two vertices u and v of G there is a (u, v)-path in G [9, p. 48]. The board states the same definition: "Graph is called connected if for any n_i, n_j there exists a path between nodes" [4, 5].

Remark 2.49. *First*, the textbook's "path" here is its Definition 1.2.12, a trail [9, p. 48]. The definition does not depend on the choice among walk, trail, and path: if a walk joins two vertices then a path joins them (Lemma 2.44), and a path is a trail (Proposition 2.35) and a walk, so the three readings accept exactly the same graphs; this is also stated in [3, Proposition 2.9.10]. *Second*, "any two vertices" includes $u = v$, and such a pair is always joined by the walk of length 0; so the definition may be read as "any two different vertices" without change. The textbook does not discuss the case $u = v$. *Third*, the definition forbids exactly one thing: a pair of vertices with no trip between them.

Example 2.50. B is connected: n_1 and n_3 are joined by the path 1-2-3, and the other pairs by one edge. The textbook's examples are the complete graphs, the complete bipartite graphs, and the three graphs of its Fig. 2.1.1 [9, p. 48].

Non-example 2.51. Three vertices with the single edge n_1n_2 are not connected: no walk leaves n_3, so the pair n_1, n_3 has no path. The graph of the textbook's Fig. 2.1.2 is not connected and has four components [9, p. 49].

In plain words. Look at a village that is not in one piece. It falls into pieces. Inside one piece every house can reach every other house. No road joins two different pieces. A village in one piece has exactly one piece.

Definition 2.52 (component; textbook p. 49, in prose, and Remark 2.1.1)**.** On V define $u \sim v$ to mean that G has a (u, v)-path. This relation is reflexive, symmetric, and transitive, so V splits into classes $V_1, \ldots, V_p$. The subgraphs $G_j = G[V_j]$, consisting of the vertices in V_j and the edges of G that join two of them, are the *connected components* of G [9, p. 49]. The textbook's Remark 2.1.1 states that G is connected if and only if it has exactly one component [9, p. 49]; see also [3, Definition 2.9.12].

Remark 2.53. A component is a whole class: a vertex that can be reached from a vertex of V_j lies in V_j.

Example 2.54. The graph with the single edge n_1n_2 on three vertices has the two components $G[\{n_1, n_2\}]$ and $G[\{n_3\}]$; B has the single component $G[\{n_1, n_2, n_3\}]$.

Non-example 2.55. $G[\{n_1\}]$ is not a component of the graph with the single edge n_1n_2, although it is connected: n_2 can be reached from n_1, so n_2 is in the same class.

3 Question 1: what the entry $(A^k)_{ij}$ represents

The answer is short and the proof is an induction. Before the proof we fix what a power of a matrix is.

3.1 Matrix products and powers

In plain words. To multiply the table of roads by itself, take one row house and one column house. Go along the row and down the column together. Multiply the two numbers you meet at each step and add up the results. Repeat for every pair of houses. The result is a new table of the same size.

Definition 3.1 (matrix product and power)**.** For $N \times N$ matrices X and Y, the *product* XY is the $N \times N$ matrix with

$$(XY)_{ij} = \sum_{r=1}^{N} X_{ir}\, Y_{rj}. \tag{1}$$

The *powers* of A are $A^1 = A$ and $A^k = A^{k-1}A$ for $k \geq 2$.

Remark 3.2. In this section the exponent k is a positive integer. The power A^0, which is the identity matrix I (ones on the diagonal, zeros elsewhere), is used only in Section 4.

Example 3.3. For the board matrix A of Definition 2.9, the entry $(A^2)_{11}$ is $0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$.

Non-example 3.4. A^2 is not the matrix of squared entries. For the board matrix, $(A^2)_{22} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$, while $A_{22}^2 = 0$.

3.2 The theorem

In plain words. Take the table of roads and multiply it by itself. Each number in the new table counts trips of two roads. The trips run from the row house to the column house. Multiply again and the numbers count trips of three roads. The count includes trips that turn back. So the table counts trips, not routes.

Theorem 3.5 (walk counting)**.** *Let A be the adjacency matrix of a simple undirected graph G with vertices $n_1, \dots, n_N$, and let $k \geq 1$ be an integer. Then $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j, for all i, j.*

This answers question Q1: the entry is a number, and it counts walks. The theorem is stated for simple graphs in [2, Theorem 1.1.3] and [1, Theorem 1], and for digraphs and multidigraphs, of which a simple undirected graph is the symmetric case, in [3, Theorem 4.5.10, pp. 120–121] and [6, Corollary 9.3.3]. The textbook does not contain it (Section 1). The proof below is written here and follows the standard route of [2, proof of Theorem 1.1.3]: induction on k, splitting the walks by their second-last vertex.

Remark 3.6. The theorem allows $i = j$, so it counts closed walks, and it counts every walk once, with repeated vertices and edges allowed. It forbids reading the number as a count of paths (Proposition 3.10).

Example 3.7. For the board graph, $(A^2)_{13} = 1$: the only walk of length 2 from n_1 to n_3 is 1-2-3 (Example 6.1).

Non-example 3.8. $(A^2)_{12} = 0$ does not say that n_1 and n_2 are unconnected. It says only that no walk of length exactly 2 joins them; walks of length 1 and of length 3 do.

Proof. Write $W_k(i, j)$ for the number of walks of length k from n_i to n_j. We prove $W_k(i, j) = (A^k)_{ij}$ for all i, j, by induction on $k \geq 1$. By Remark 2.20 we count vertex sequences $w_0, \dots, w_k$ with $w_0 = n_i$, $w_k = n_j$, and w_{t-1} adjacent to w_t for each t.

Base case $k = 1$. A walk of length 1 from n_i to n_j is a sequence (n_i, n_j) with n_i and n_j adjacent. There is exactly one such sequence if they are adjacent and none otherwise. So $W_1(i, j) = A_{ij} = (A^1)_{ij}$.

Induction hypothesis. Let $k \geq 2$ and suppose $W_{k-1}(i, j) = (A^{k-1})_{ij}$ for all i, j.

Inductive step. Fix i and j. Every walk $w_0, \dots, w_k$ of length k from n_i to n_j has a second-last vertex w_{k-1}. Write $w_{k-1} = n_r$ with $r \in \{1, \dots, N\}$ and sort the walks into N classes by the value of r. Each walk has exactly one second-last vertex, so the classes are disjoint and together contain every walk:

$$W_k(i, j) = \sum_{r=1}^{N} \big(\text{number of walks of length } k \text{ from } n_i \text{ to } n_j \text{ with second-last vertex } n_r\big).$$

We claim that class r has exactly $W_{k-1}(i, r)\, A_{rj}$ members.

- If n_r and n_j are not adjacent, then $A_{rj} = 0$, and no walk can have n_r followed by n_j, so the class is empty.
- If n_r and n_j are adjacent, then $A_{rj} = 1$. Deleting the last vertex of a walk in the class leaves $w_0, \dots, w_{k-1}$, a walk of length $k - 1 \geq 1$ from n_i to n_r. Conversely, appending n_j to any walk of length $k - 1$ from n_i to n_r gives a walk of length k from n_i to n_j, because the new pair n_r, n_j is adjacent, and its second-last vertex is n_r. Deleting and appending undo each other, so they match the class one to one with the walks of length $k - 1$ from n_i to n_r, and the class has $W_{k-1}(i, r)$ members.

Therefore $W_k(i, j) = \sum_r W_{k-1}(i, r) A_{rj}$. By the induction hypothesis this equals $\sum_r (A^{k-1})_{ir} A_{rj}$, which is $(A^{k-1}A)_{ij} = (A^k)_{ij}$ by (1). This proves the statement for k, and by induction for all $k \geq 1$. □

Why the argument is valid. Counting could go wrong in two places: a walk counted twice or missed, and a count that is not 0 or 1 where adjacency is concerned. Sorting by the second-last vertex counts each walk exactly once, and the deleting-and-appending match is one to one. Simplicity is used twice: it identifies a walk with its vertex sequence (Remark 2.20), and it makes A_{rj}, the number of edges between n_r and n_j, equal to 0 or 1. Undirectedness is not used in the induction itself; it only makes A symmetric, which gives part (c) of the next corollary. For a multigraph the same argument would count walks with multiplicity if A_{rj} were the number of edges, which agrees with the sources that state the theorem for multidigraphs [3, Theorem 4.5.10].

3.3 Consequences

In plain words. A count is never negative and always a whole number. The table of trips looks the same when flipped over the line of squares from top left to bottom right. That is because every trip can be driven backwards. A positive number means that at least one such trip exists. A trip of two roads from a house back to itself goes out along a road. Then it comes back along the same road.

Corollary 3.9 (consequences of walk counting)**.** *For a simple undirected graph and $k \geq 1$:*

(a) every entry of A^k is a nonnegative integer, and a sum of powers of A has a positive (i, j) entry if and only if some summand does;

(b) $(A^k)_{ij} > 0$ if and only if there is a walk of length exactly k from n_i to n_j;

(c) A^k is symmetric;

(d) $(A^2)_{ii} = \deg(n_i)$ for every i, hence $\operatorname{trace}(A^2) = 2|E|$.

Proof. Parts (a) to (c) are derived here from Theorem 3.5; part (d) is stated in [2, remark after Theorem 1.1.3]. (a) and (b): by Theorem 3.5 the entry is a count; a sum of nonnegative numbers is positive if and only if one of them is. (c): reversing the vertex sequence of a walk matches the walks from n_i to n_j one to one with the walks from n_j to n_i, because edges are unordered; so $(A^k)_{ij} = (A^k)_{ji}$. (d): a walk of length 2 from n_i to n_i has the form n_i, n_r, n_i with n_r a neighbour of n_i, and there is one for each neighbour, so $(A^2)_{ii}$ is the number of neighbours, which is $\deg(n_i)$. The trace is the sum of the diagonal, so it is the sum of the degrees, which is $2|E|$ by Proposition 2.18. □

3.4 The warning: walks are not paths

In plain words. A trip may turn back and pass the same house twice. A route never passes a house twice. The table counts trips, so it counts the turn-back trips too. A house with a neighbour always has a two-road trip that goes out and comes back. That trip is not a route. So the table cannot be counting routes.

Proposition 3.10 (the entries do not count paths)**.** *(a) If n_i has at least one neighbour, then $(A^2)_{ii} = \deg(n_i) > 0$, but there is no trail and no path from n_i to n_i of length 2.*

(b) In the board graph, $(A^3)_{12} = 2$, but the only path or trail from n_1 to n_2 has length 1. More fully, Table 2 lists, for $k = 1, \ldots, 5$, the entry $(A^k)_{12}$ and the number of trails and of paths among the walks it counts.

In particular the entries of A^k are not counts of paths, and not counts of trails.

Walks, not paths, are counted also according to [1, Section 1, remark after Theorem 1] and [3, Example 2.9.2], which gives $1, 2, 1$ as a walk that is not a path. The proposition and its proof are written here.

Proof. (a) The first claim is Corollary 3.9(d). The walks of length 2 from n_i to n_i are the sequences n_i, n_r, n_i. Each repeats the vertex n_i, so it is not a path (Definition 2.31), and each uses the edge $n_i n_r$ twice, so it is not a trail (Definition 2.27). Under the course convention the only path from n_i to n_i is the single vertex, of length 0 (Remark 2.32).

(b) The walks counted by $(A^3)_{12}$ are 1-2-1-2 and 1-2-3-2 (Example 6.1); the first repeats the edge $n_1 n_2$ and the second repeats the vertex n_2, so neither is a trail or a path. Every path in B has at most $N - 1 = 2$ edges (Corollary 2.46); a path of length 2 from n_1 to n_2 would need a vertex adjacent to both, and there is none, so the only path from n_1 to n_2 is the edge. A trail has at most $|E| = 2$ edges, and no walk of length 2 joins n_1 to n_2 because $(A^2)_{12} = 0$, so the only trail from n_1 to n_2 is also the edge. The rows of Table 2 come from an enumeration by script (Appendix D); the walk counts in them equal the entries of A^k computed by matrix multiplication. □

Table 2: Walks, trails, and paths from n_1 to n_2 in the board graph, by length k. The second column is $(A^k)_{12}$.

k	walks $= (A^k)_{12}$	trails among them	paths among them
1	1	1	1
2	0	0	0
3	2	0	0
4	0	0	0
5	4	0	0

What a positive entry does guarantee. If $(A^k)_{ij} > 0$ then a walk of length k exists (Corollary 3.9(b)), and then a path of length at most k exists (Lemma 2.44). So the warning does no harm to the connectivity test of Section 4, but the value of an entry is not a count of paths. The mistake of reading the entry as a count of paths is made in one of the sources (Section 7.4).

4 Question 2: deciding connectivity from A

Connectedness is defined by paths (Definition 2.48), but powers of A count walks (Theorem 3.5). The bridge is Lemma 2.44: a walk can always be cut down to a path. The textbook states no matrix test for connectivity (Section 1). The test below is stated in [7, Theorem 2] and [11], and in the form with $(I + A)^{N-1}$ in [1, Theorem 2]. The two web sources that give a proof prove it loosely, speaking of "paths" where walks are meant (Section 7.4), so the proof is written out here. Throughout, G has $N \geq 2$ vertices; Remark 4.9 treats $N = 1$.

4.1 The sum of the first $N - 1$ powers

In plain words. We need one table that gathers all the short trips. For each pair of houses, count the trips of one road, of two roads, and so on. Stop at one road fewer than the number of houses. Add the counts for each pair. Call the result the table of totals.

Definition 4.1 (the sum S)**.** For a graph with $N \geq 2$ vertices, let

$$S = A + A^2 + \cdots + A^{N-1}, \tag{2}$$

the sum of the first $N-1$ powers of A, added entry by entry. This notation is introduced here and is not the textbook's.

Remark 4.2. By Theorem 3.5, S_{ij} is the number of walks from n_i to n_j whose length is between 1 and $N-1$; so every entry of S is a nonnegative integer (Corollary 3.9(a)). The definition needs $N \geq 2$, since for $N = 1$ the sum would be empty.

Example 4.3. For the board graph ($N = 3$), $S = A + A^2$ and its first row is 1 1 1 (Example 6.2).

Non-example 4.4. A^2 alone is not S for $N = 3$, and neither is A alone. For the board graph A^2 has the zero entry $(1, 2)$ although n_1 and n_2 are joined by an edge, and A has the zero entry $(1, 3)$ although n_1 and n_3 are connected; either matrix alone would wrongly report "not connected".

4.2 The test and its proof

In plain words. We want a quick check for a village. We only have the table of roads. From the table we can count trips. Count the trips with one road, two roads, and so on, for each pair of houses. If no pair of different houses has a total of zero, the village is in one piece. We never need trips longer than one road fewer than the number of houses. A trip that wastes time can always be shortened.

Proposition 4.5 (one pair at a time)**.** *Let G have $N \geq 2$ vertices and let $i \neq j$. The following three statements are equivalent.*

(i) There is a path between n_i and n_j (Definition 2.31).

(ii) There is an integer k with $1 \leq k \leq N-1$ and $(A^k)_{ij} > 0$.

(iii) $S_{ij} > 0$.

Theorem 4.6 (the connectivity test)**.** *Let G be a graph with $N \geq 2$ vertices and adjacency matrix A. The following are equivalent.*

(a) G is connected.

(b) Every entry off the diagonal of $S = A + A^2 + \cdots + A^{N-1}$ is positive.

(c) Every entry of $(I + A)^{N-1}$ is positive.

This answers question Q2. The equivalence of (a) and (b) is stated in [7, Theorem 2]; the equivalence of (a) and (c) in [1, Theorem 2]. The proofs below are written here.

Proof of Proposition 4.5. (i) ⇒ (ii). A path between n_i and n_j is a walk, say of length ℓ. By Corollary 2.46 there is a walk from n_i to n_j of length $k \leq N-1$. Since $n_i \neq n_j$ the walk has at least one step, so $k \geq 1$. By Theorem 3.5, $(A^k)_{ij} \geq 1$.

(ii) ⇒ (iii). Each term of $S_{ij} = A_{ij} + (A^2)_{ij} + \cdots + (A^{N-1})_{ij}$ is a nonnegative integer (Corollary 3.9(a)), and the term with index k is positive, so the sum is positive.

(iii) ⇒ (ii). A sum of nonnegative terms is positive only if some term is positive.

(ii) ⇒ (i). If $(A^k)_{ij} > 0$ then, by Theorem 3.5, some walk of length k joins n_i to n_j. By Lemma 2.44 there is a path between n_i and n_j. □

Proof of Theorem 4.6. Statement (a) says that for every pair of vertices there is a path (in the sense of Definition 2.48). The pairs with $u = v$ are always joined by the walk of length 0 (Remark 2.49), so (a) holds if and only if a path joins n_i and n_j for every $i \neq j$ (by Remark 2.49 the reading of "path" does not matter). By Proposition 4.5 this holds if and only if $S_{ij} > 0$ for every $i \neq j$, which is (b). By Proposition 4.17 with $m = N - 1$, it holds if and only if every off-diagonal entry of $(I + A)^{N-1}$ is positive; the diagonal entries are always positive there; so it holds if and only if (c). □

Corollary 4.7 (components from S)**.** *For any graph, connected or not, with $N \geq 2$, and $i \neq j$: $S_{ij} > 0$ if and only if n_i and n_j lie in the same component.*

Proof. Proposition 4.5 does not use the connectivity of G; statement (i) says $n_i \sim n_j$, which is the definition of lying in the same class (Definition 2.52). □

Remark 4.8 (by-product: distance). If n_i and n_j are in the same component and $i \neq j$, then $d(n_i, n_j)$ is the smallest $k \geq 1$ with $(A^k)_{ij} > 0$, and this k is at most $N - 1$. Indeed $(A^k)_{ij} > 0$ exactly when a walk of length k exists (Corollary 3.9(b)), and the length of a shortest walk is the distance (Proposition 2.47). Two sources state this [6, Section 9.3] [7, Theorem 3].

Remark 4.9 (the diagonal, and $N = 1$, $N = 2$). The test looks only at entries off the diagonal. For $N = 2$ and one edge, $S = A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has zeros on the diagonal although the graph is connected, so the diagonal of S cannot be required to be positive. For $N \geq 3$ and G connected the diagonal of S is positive anyway: every vertex has a neighbour, and $(A^2)_{ii} = \deg(n_i) > 0$ by Corollary 3.9(d). For $N = 1$ there is no pair of different vertices, G is connected, and $(I + A)^0 = I = [1]$ has no zero entry; the sum S is empty, so the test is stated for $N \geq 2$.

4.3 The test as a procedure

In plain words. Here is the check, step by step. First write the table of roads. Then multiply the table by itself again and again. After each step, add the new counts to a running total. Stop one step before the number of houses. Look at every pair of different houses. If any total is zero, those two houses cannot reach each other. If no total is zero, everyone can reach everyone.

Test 4.10 (matrix test for connectivity)**.** *Input:* the adjacency matrix A of a graph with $N \geq 2$ vertices.

1. Set $P = A$ and $S = A$.
2. For $k = 2, 3, \ldots, N - 1$: replace P by PA (now $P = A^k$) and add P to S.
3. Look at the entries S_{ij} with $i \neq j$. If one of them is 0, the graph is not connected; if none is 0, it is connected.

Equivalent form: compute $(I + A)^{N-1}$ and report "connected" if it has no zero entry.

Remark 4.11. The test allows any graph with $N \geq 2$ vertices. It forbids stopping early (Proposition 4.14) and forbids applying it to a directed graph, which this note does not treat. A zero at (i, j) means, by Corollary 4.7, that n_i and n_j are in different components.

Example 4.12. The board matrix gives "connected" and the matrix with the single edge $n_1 n_2$ gives "not connected"; both are worked in Examples 6.2 and 6.3.

Non-example 4.13. Computing only A^{N-1} is not the test: for the board graph A^2 has the zero entry $(1, 2)$ although the graph is connected.

4.4 Why $N-1$ is enough and $N-2$ is not

Enough. If a walk between two different vertices exists at all, then a walk of length at most $N-1$ exists (Corollary 2.46), because a shortest connection repeats no vertex and a repeat-free connection has at most N vertices. Adding powers beyond $N-1$ cannot change which off-diagonal entries are positive.

In plain words. Take four houses in a row. To go from the first house to the last, you must pass every house in between. That takes three roads. If we counted only trips of up to two roads, we would miss this pair. We would say the village is cut in two, and we would be wrong. The same idea works for any row of houses.

Proposition 4.14 ($N-1$ cannot be lowered to $N-2$)**.** *For every $N \geq 3$ there is a connected graph with N vertices for which the sum $A + \cdots + A^{N-2}$ has a zero entry off the diagonal.*

Proof. This is proved here. Take the graph with edges $n_1n_2, n_2n_3, \ldots, n_{N-1}n_N$. Each step of a walk changes the vertex number by exactly 1, and to go from n_1 to n_N the number must increase by $N-1$, so every walk from n_1 to n_N has length at least $N-1$. By Theorem 3.5, $(A^k)_{1N} = 0$ for $1 \leq k \leq N-2$, so the $(1,N)$ entry of $A + \cdots + A^{N-2}$ is 0, although the graph is connected. □

Example 4.15. Path graph on four vertices with edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$):

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 2 & 0 \end{pmatrix},$$

$$A + A^2 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad A + A^2 + A^3 = \begin{pmatrix} 1 & 3 & 1 & 1 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 1 & 1 & 3 & 1 \end{pmatrix}.$$

With $N-2 = 2$ powers the $(1,4)$ entry is 0 and the test would wrongly say "not connected"; with $N-1 = 3$ powers every entry is positive.

Table 3: The path graph on N vertices: the smallest k with $(A^k)_{1N} > 0$, and the $(1,N)$ entry of the sum with $N-2$ powers and with $N-1$ powers.

N	smallest k with $(A^k)_{1N} > 0$	$(1,N)$ of $A + \cdots + A^{N-2}$	$(1,N)$ of $A + \cdots + A^{N-1}$
3	2	0	1
4	3	0	1
5	4	0	1
6	5	0	1
7	6	0	1
8	7	0	1

Remark 4.16. Which graphs fail with $N-2$ powers? By Remark 4.8 a connected graph fails exactly when some pair has distance $N-1$. A shortest path with $N-1$ edges contains all N vertices, and an edge between two vertices of it that are not consecutive would shorten the distance between its ends; so the graph is that path and nothing else. The failing graphs are therefore exactly the labelled paths, of which there are $N!/2$ for $N \geq 3$: $3, 12, 60, 360$ for $N = 3, 4, 5, 6$. The enumeration of Table 6 gives the same four numbers.

4.5 The form $(I + A)^m$

In plain words. There is a second form of the check. Let every house also stay where it is for one step. Then count again. The answer is the same. You may even go on past the number of houses minus one. The pattern of zeros does not change.

Proposition 4.17 (powers of $I + A$)**.** *Let $N \geq 2$ and let $m \geq N - 1$ be an integer. Every diagonal entry of $(I + A)^m$ is at least 1, and for $i \neq j$ the entry $((I + A)^m)_{ij}$ is positive if and only if there is a path between n_i and n_j. Hence G is connected if and only if $(I + A)^m$ has no zero entry, for any one such m, and an exponent $m = 2^t \geq N - 1$ is reached by t squarings of $I + A$.*

The case $m = N - 1$ is the form stated in [1, Theorem 2]. The proof, by binomial expansion, is written in Appendix C.4.

4.6 An alternative: breadth-first search

In plain words. There is another way, and it needs no counting. Stand at one house and mark it. Then mark every house next to it. Then mark every house next to those. Keep going until no new house appears. If every house is marked, the village is in one piece.

Remark 4.18 (breadth-first search)*.* Breadth-first search from a start vertex s keeps a queue of discovered vertices and repeatedly takes the vertex at the front and discovers its undiscovered neighbours; its correctness statement is that for every vertex v there is a walk from s to v if and only if v is discovered when the search ends, so G is connected if and only if every vertex is discovered from one start vertex [10, slides 6 and 10]. When the graph is given by adjacency lists the search takes time proportional to the number of vertices plus the number of edges [10, slides 9 and 10]. It gives the same answer as the matrix test on every labelled graph with at most six vertices (Table 6), a computer check and not a proof, and it lists the components directly.

5 Question 3: a path between two vertices, and the three usages of the word

The answer to question Q3 is Definition 2.31, which sits in its place in the textbook's build order: a path between n_i and n_j is a walk from n_i to n_j whose vertices are pairwise distinct. This section adds only what is new there: how that word compares with the other usages, and what depends on the choice. The course meaning of "path" (no repeated vertex) comes from the assignment text as reported to us (Item 4); the textbook supplies the definition under another name, and the board does not say which meaning it uses [4, 5].

5.1 Three usages

The same two properties of a walk, "no repeated edge" and "no repeated vertex", get different names in different places. In the textbook, "path" means no repeated edge [9, Definition 1.2.12, p. 22] and "simple path" means no repeated vertex [9, Definition 1.2.13, p. 22]. In the English text used here, a path has distinct vertices and a trail has distinct edges [3, Definitions 2.9.1(c) and 3.4.5]; an encyclopedia article gives the same two definitions and adds that a path has distinct vertices "by most definitions" [8]. The same article records that some authors do not require distinct vertices in a path and say "simple path" for a path with distinct vertices [8], which is the textbook's pattern; so the course convention is one of at least two in use. For "cycle" the textbook's word (Definition 2.36) allows repeated vertices, while the English text defines a cycle as a closed walk with at least three steps whose vertices $w_0, \ldots, w_{k-1}$ are distinct

[3, Definition 2.10.1(b)], which is the textbook's simple cycle. Table 4 puts the usages side by side.

Table 4: The naming of walks in the textbook, in English usage, and in this note. The textbook words are given in English translation.

Property of the walk	Textbook (definition number)	English usage	This note
any sequence of steps along edges	walk (1.2.10)	walk	walk
first vertex equals last vertex	closed walk (1.2.11)	closed walk	closed walk
no repeated edge, vertices may repeat	"path" (1.2.12)	trail	trail
no repeated vertex (hence no repeated edge)	"simple path" (1.2.13)	path	**path**
closed, no repeated edge, vertices may repeat	"cycle" or "closed path" (1.2.14)	closed trail	cycle (textbook sense)
closed, only first and last vertex repeat, at least three steps	"simple cycle" (1.2.15)	cycle	simple cycle

The word "path" therefore has three meanings across the sources: no repeated edge (textbook), no repeated vertex (English text and course), and, in some authors, either. The last column of Table 4 gives each meaning one name in this note and is used without exception.

5.2 What depends on the choice and what does not

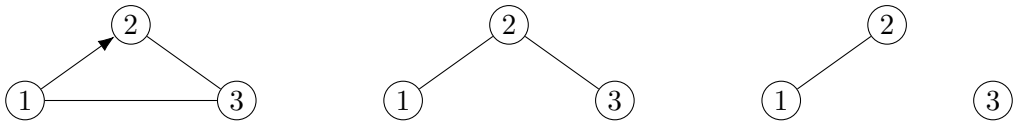
Existence does not depend on it. For two different vertices, a walk, a trail, and a path between them exist together or not at all: a walk contains a path (Lemma 2.44), and a path is a trail (Proposition 2.35) and a walk. So every statement of the form "there is a path between n_i and n_j" has the same truth value under the textbook's reading and under the course reading, and so do connectivity (Remark 2.49) and distance (Proposition 2.47). The board's definition of connected is therefore the textbook's Definition 2.1.1 under either reading.

Counts do depend on it. The number of walks, of trails, and of paths from n_i to n_j are different numbers in general, and only the first is given by a power of A (Proposition 3.10). In the board graph every trail is a path (Table 5), so that graph cannot show the difference between the two readings; it needs a graph with a cycle, such as the textbook's own example graph (Definition 2.27).

Length zero. By Item 6 a single vertex is a path of length 0; this matches the English text, which calls (5) a walk of length 0 and a path [3, Example 2.9.2]. It matters only for $i = j$, since for $i \neq j$ a path has length at least 1.

6 Worked examples

This section works the board graph through Section 3 and Section 4, and adds one graph that is not connected. Every matrix, sum, and count in this section and in earlier examples was computed by a script (Appendix D) and then copied into the text by the same script.



(a) board drawing (b) board graph B (c) one edge only

Figure 1: (a) The board drawing as read from the photographs: a line from 1 to 2 with an arrowhead at 2, a line from 1 to 3, and a line from 2 to 3. (b) The graph B whose adjacency matrix is the board matrix: the edges n_1n_2 and n_2n_3 only. (c) The graph with the single edge n_1n_2, used as the example that is not connected. The positions of the vertices are our choice.

6.1 The powers of the board matrix

In plain words. We take the village from the board. It has three houses in a row. The first and last houses each have one road. The middle house has two roads. We multiply the table of roads by itself several times. Then we list every trip, so that the reader can see that each number is right.

Example 6.1 (powers of the board matrix)**.** The graph is B (Figure 1(b)) with

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

$$A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \qquad A^5 = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

Reading of A^2 and A^3 by walks. The walks of length 2 are: from n_1 to n_1, 1-2-1; from n_1 to n_3, 1-2-3; from n_2 to n_2, 2-1-2, 2-3-2; from n_3 to n_3, 3-2-3. The walks of length 3 are: from n_1 to n_2, 1-2-1-2, 1-2-3-2; from n_2 to n_3, 2-1-2-3, 2-3-2-3. Each count equals the corresponding matrix entry (Theorem 3.5).

The induction step by hand. $(A^2)_{22} = \sum_r A_{2r}A_{r2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$. $(A^3)_{12} = \sum_r (A^2)_{1r}A_{r2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$, where the term $r = 1$ is the walk 1-2-1 extended by a step to n_2, giving 1-2-1-2, and the term $r = 3$ is the walk 1-2-3 extended by a step to n_2, giving 1-2-3-2.

Patterns. The diagonal of A^2 is $(1, 2, 1)$, the degrees, and $\text{trace}(A^2) = 4 = 2|E|$ (Corollary 3.9(d)). Also $A^3 = 2A$, and therefore $A^{k+2} = A^{k-1}A^3 = 2A^{k-1}A = 2A^k$ for every $k \geq 1$; the matrices above agree with this for $k = 1, 2, 3$.

Table 5: Walks, trails, paths, and closed walks of each length in the board graph, counted over all pairs of start and end vertices by enumeration. For $k \geq 1$ the number of walks is the sum of all entries of A^k.

length k	walks	trails	paths	closed walks
0	3	3	3	3
1	4	4	4	0
2	6	2	2	4
3	8	0	0	0
4	12	0	0	8

Table 5 shows the effect of Proposition 3.10: the number of walks grows with k, while trails and paths stop at length 2, because B has two edges and $N-1=2$. It also shows that in B every trail is a path: B has no cycle (Definition 2.36), and a trail that repeated a vertex would contain a closed trail between the two visits.

6.2 A connected graph

In plain words. Take three houses in a row. A road joins the first and second house. Another road joins the second and third house. The first and third house share no road. Can you still go from one to the other? Yes, through the middle house. We expect the check to say that the village is in one piece.

Example 6.2 (connected: the board graph, $N=3$, $N-1=2$)**.**

$$A=\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix},\quad A^2=\begin{pmatrix}1&0&1\\0&2&0\\1&0&1\end{pmatrix},\quad S=A+A^2=\begin{pmatrix}1&1&1\\1&2&1\\1&1&1\end{pmatrix},\quad (I+A)^2=\begin{pmatrix}2&2&1\\2&3&2\\1&2&2\end{pmatrix}.$$

S has no zero entry (in particular none off the diagonal), so by Theorem 4.6(b) the graph is connected. $(I+A)^2$ has no zero entry, so by Theorem 4.6(c) the graph is connected; the two answers agree. The entry A_{13} is 0, so no edge joins n_1 and n_3, yet $(A^2)_{13}=1$: exactly one walk of length 2 joins them, namely 1-2-3. The entry $((I+A)^2)_{13}=1=\binom{2}{2}(A^2)_{13}$, since the other two terms of the expansion vanish there; and $(I+A)^2=I+2A+A^2$ holds entry by entry, with $I=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$ and $2A=\begin{pmatrix}0&2&0\\2&0&2\\0&2&0\end{pmatrix}$. Breadth-first search from n_1 reaches n_1, n_2, n_3, the single component, and agrees.

6.3 A graph that is not connected

In plain words. Now take three houses again. Only one road exists, between the first and the second house. The third house stands alone. Nobody can drive to it. We expect the check to say that the village is not in one piece. We also expect it to show that the third house is a piece by itself.

Example 6.3 (not connected: edge n_1n_2 only, $N=3$, $N-1=2$)**.**

$$A=\begin{pmatrix}0&1&0\\1&0&0\\0&0&0\end{pmatrix},\quad A^2=\begin{pmatrix}1&0&0\\0&1&0\\0&0&0\end{pmatrix},\quad S=A+A^2=\begin{pmatrix}1&1&0\\1&1&0\\0&0&0\end{pmatrix},\quad (I+A)^2=\begin{pmatrix}2&2&0\\2&2&0\\0&0&1\end{pmatrix}.$$

The entries off the diagonal of S that are 0 are at $(1,3)$, $(2,3)$, $(3,1)$, and $(3,2)$, and the zeros of $(I+A)^2$ are at the same four places. By Theorem 4.6 the graph is not connected, and by Corollary 4.7 the vertices n_1 and n_2 lie in one component and n_3 in another: the components are $G[\{n_1,n_2\}]$ and $G[\{n_3\}]$. Breadth-first search from n_1 reaches only n_1 and n_2, and agrees. Higher powers do not help: the entries $(1,3)$ and $(2,3)$ of A^k are 0 for $k=1,\dots,6$, and in fact no walk reaches n_3 at all.

6.4 A computer check over all small graphs

Table 6 reports a check, not a proof. For every labelled simple graph on $N=2,\dots,6$ vertices, the sum test, the $(I+A)^{N-1}$ test, and the repeated-squaring form of Proposition 4.17 were compared with an independent breadth-first search; the table counts the graphs on which each

agrees. The last column counts the graphs on which the sum with only $N-2$ powers gives the wrong answer (for $N=2$ that sum is empty). The component statement of Corollary 4.7 was checked on the same graphs by a second script (Appendix D).

Table 6: All labelled simple graphs on N vertices. Columns 4 to 6 count the graphs on which the named test agrees with breadth-first search.

N	graphs	connected	sum test	$(I+A)^{N-1}$	squaring	$N-2$ powers wrong
2	2	1	2	2	2	1
3	8	4	8	8	8	3
4	64	38	64	64	64	12
5	1024	728	1024	1024	1024	60
6	32768	26704	32768	32768	32768	360

7 Discussion: readings of the board, limits, and competing explanations

7.1 Where the board and the matrix disagree

The drawing on the board has three lines and the matrix records two edges [4, 5]. Edge by edge: the pair n_1, n_2 is drawn (with an arrowhead at n_2) and has $A_{12} = A_{21} = 1$; the pair n_2, n_3 is drawn and has $A_{23} = A_{32} = 1$; the pair n_1, n_3 is drawn but has $A_{13} = A_{31} = 0$ (Figure 1). The matrix is symmetric, so it gives no support to a direction on the line from n_1 to n_2. Following Item 3, this note takes the matrix as the example, because the board says that the graph is given through its adjacency matrix [4]. If the drawing had been meant, the adjacency matrix would be $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, which differs from the board matrix exactly at the entries $(1,3)$ and $(3,1)$. This note does not decide which of the two the lecturer intended. The choice does not affect the answers to Q1 to Q3, which are theorems for every simple undirected graph; it changes only the numbers in the examples. Both graphs are connected, so the test of Theorem 4.6 answers "connected" for both.

7.2 The mark above A

In both photographs the mark above A in question Q1 resembles a small n or a pair of quotation marks, not clearly a k [4, 5]. The board's own letter for the number of vertices is the capital N, so a lower-case n as an exponent would be unusual, but it cannot be ruled out. Item 2 reads the mark as k because the assignment text asks about A^k. Since Theorem 3.5 holds for every positive integer exponent, it answers the board question whichever letter was written, including an exponent equal to N.

7.3 Limits of the results

- *Scope.* The theorems are proved for finite simple undirected graphs. For directed graphs and multigraphs, the walk-counting statement is made in other sources [3, Theorem 4.5.10][6, Corollary 9.3.3], and this note makes no claim about the connectivity test there.
- *What the entries do not say.* The entries count walks; they say nothing about the number of paths, which needs other methods and is not treated here (Proposition 3.10).

- *What the test does not say.* It answers "connected" or "not connected"; it does not list a path between two vertices. Corollary 4.7 gives the components and Remark 4.8 the distances.
- *Finite checks.* The computer checks of Table 6 cover graphs with at most six vertices; they support the statements but the proofs are the arguments of Section 4.2 and Appendix C.
- *Sources.* The textbook was searched in full text for the adjacency matrix and read page by page only where listed in Appendix D, so the statement that it contains neither result is a statement about the search. Several sources were read only in part, as Appendix D records.

7.4 Competing explanations

Reading the entry as a count of paths. Theorem 1 of one source says that $(A^l)_{ij}$ is the number of "different paths" of length l [7, Theorem 1]. As written it is wrong for any definition of path that forbids repeated vertices or repeated edges, by Proposition 3.10: $(A^2)_{ii}$ is positive although no such path exists. The same paper states, in its property 7, that the entry equals the number of walks [7, property 7]. This note therefore does not use that Theorem 1 as support for Theorem 3.5; the four sources cited there state walks.

Explaining the test by paths. The proof of the test in the same paper applies its Theorem 1 to $A + \cdots + A^{N-1}$ and so counts "paths" where walks are meant, and it passes from a nonzero entry to the existence of a path without the walk-to-path step [7, proof of Theorem 2]. The proof sketch of the $(I + A)^{N-1}$ form likewise says that the power "aggregates all paths" [1, Section 1, Theorem 2, proof sketch]. Both *statements* of the test are correct, as Theorem 4.6 shows; the explanations are incomplete, which is why the proof is written out in Section 4.2.

A different method. Breadth-first search gives the same answer (Remark 4.18); eigenvalue methods are outside the scope of this note and no claim is made about them.

7.5 Cost

This remark is an estimate that counts scalar multiplications with the school method, which uses N^3 of them for one product of two $N \times N$ matrices; faster methods were not examined. Reaching A^{N-1} takes $N-2$ products, which is 98,000,000 multiplications for $N = 100$. Because Proposition 4.17 allows any exponent $m \geq N-1$, squaring $I + A$ repeatedly until the exponent is at least $N-1$ takes 7 squarings, or 7,000,000 multiplications, for $N = 100$. Breadth-first search with adjacency lists takes time proportional to the number of vertices plus the number of edges [10, slides 9 and 10]; when the input is the matrix A itself it reads each of the $N^2 = 10{,}000$ entries once [10, slides 3 and 6]. The entries of the powers also grow quickly: for the complete graph on 10 vertices the largest entry of A^9 is 38,742,049. The matrix test is useful because it is a clean statement about powers of A, not because it is fast.

7.6 What this note does not settle

The board shows the question "How to know if graph is connected or not" but not its answer [4, 5], so this note makes no claim about whether the lecturer's intended answer was the sum test, the $(I + A)^{N-1}$ form, or a search.

8 Conclusion

Q1. For a finite simple undirected graph with adjacency matrix A and an integer $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j (Theorem 3.5, proved by induction on k). It counts walks, in which vertices and edges may repeat, and not paths: for example

$(A^2)_{ii}$ equals the degree of n_i, which is positive whenever n_i has a neighbour, although no path of length 2 returns to n_i (Proposition 3.10).

Q2. For $N \geq 2$ vertices, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive, and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 4.6). The proof uses the walk-counting theorem and the fact that a walk contains a path no longer than itself (Lemma 2.44); the last power $N - 1$ cannot be lowered to $N - 2$ (Proposition 4.14). The board matrix gives "connected" and the matrix with the single edge $n_1 n_2$ gives "not connected" (Examples 6.2 and 6.3). Breadth-first search is an alternative (Remark 4.18).

Q3. Under the course convention, a path between n_i and n_j is a walk from n_i to n_j whose vertices are pairwise distinct (Definition 2.31); this is the textbook's simple path, Definition 1.2.13 [9, p. 22]. The textbook's own word "path" (Definition 1.2.12) means a walk with distinct edges, called a trail here; the existence of a connection does not depend on the choice, but the counts do (Section 5).

The assumptions on which these answers rest are listed in Section 1.3: the mark above A read as an exponent, the matrix rather than the drawing as the example, and the course meaning of "path".

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[4] *Lecture whiteboard photo 1.* Local file sources/whiteboard-1.jpg. Photograph of a lecture board, supplied by the user. Author and date unknown. n.d.

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A Textbook numbers and board symbols

Table 7 lists, in the textbook's build order, each concept used in this note with the textbook's number and printed page, and the word the textbook uses where it differs from English usage [9]. The textbook's words are in Armenian script, which this note does not typeset; its words are given in English translation. Printed page numbers are one lower than PDF page numbers (Item 7).

Table 7: Concepts in the textbook's build order, with the textbook's numbers.

Concept	Textbook number	Printed page	Textbook word (translated)
graph $G = (V, E)$	Definition 1.1.1	9	
adjacent	Definition 1.1.4	11	
adjacency matrix $A(G)$	no number of its own, after Definition 1.1.6	11	
degree	Definition 1.2.1	16	
sum of all degrees equals $2\|E\|$	Theorem 1.2.1 (Euler)	16	
walk	Definition 1.2.10	21	
closed walk	Definition 1.2.11	22	
walk with no repeated edge (trail)	Definition 1.2.12	22	"path"
walk with no repeated vertex (path)	Definition 1.2.13	22	"simple path"
cycle	Definition 1.2.14	22	"cycle", "closed path"
simple cycle	Definition 1.2.15	22	"simple cycle"
distance	Definition 1.2.16	22–23	
every (u, v)-walk contains a simple (u, v)-path, $u \neq v$	Lemma 1.2.1	23	
connected	Definition 2.1.1	48	
connected components	in prose, no number	49	
connected if and only if one component	Remark 2.1.1	49	

Table 8 compares the board's symbols with the textbook's and with this note's. The first row records that the board's V in the line defining A_{ij} is the set of edges [4, 5].

Table 8: Board symbols, textbook symbols, and the symbols of this note.

Board	Textbook	This note
(G, V), with V the edge set	$G = (V, E)$	$G = (V, E)$
$e_{ij} \in V$	$uv \in E$	$n_i n_j \in E$ or $e_{ij} \in E$
nodes n_i	vertices v_i	vertices n_i
$\|G\| = N$ nodes	n vertices	N vertices
A_{ij}	a_{ij} in $A(G)$	A_{ij} in A

B Transcription of the whiteboard

The two photographs show the same board at slightly different angles, and they are read together [4, 5]. The lines below follow the reading order; the board's wording and spelling are kept.

1. The matrix entry is defined as $A_{ij} = 1$ if $e_{ij} \in V$ and $A_{ij} = 0$ if $e_{ij} \notin V$ [4, 5]. The board's V here is the set of edges (Item 5).
2. A drawing with vertices labelled 1, 2, 3, and next to it a grid labelled "adj." with rows 0 1 0, 1 0 1, 0 1 0 [4, 5]. The drawing shows a line from 1 to 2 with an arrowhead at 2, a line from 1 to 3, and a line from 2 to 3; the matrix records only the edges 1–2 and 2–3 (Section 7.1).
3. "Problem. Given graph (G, V) with N nodes i.e. $|G| = N$ and the graph is represented through it's adj. matrix. So you have 'A' matrix." [4, 5].
4. "1) What does $A_{ij}^{\cdot}$ represent", where $\cdot$ marks a small mark that looks like a small n or quotation marks [4, 5] (Section 7.2).
5. "2) How to know if graph is connected or not." [4, 5].
6. "Def. Graph is called connected if for any n_i, n_j there exists a path between nodes." [4, 5].

Clearly visible: the prose of items 1, 3, 5 and 6, the matrix entries, and the vertex labels. *Likely interpretation:* the mark in item 4 is an exponent k (Item 2); the arrowhead in item 2 is a stray mark (Item 3). *Not settled:* which of the drawing and the matrix the lecturer intended, and which meaning of "path" the board uses (Item 4).

C Derivations

This appendix holds the arguments that the main text only states, in the order in which the main text needs them.

C.1 Walk counting: walks and vertex sequences

The proof of Theorem 3.5 counts vertex sequences instead of walks. This is allowed by the following fact, which is stated before the theorem in Remark 2.20.

Proposition C.1. *In a simple graph, the map that sends a walk $u_0, e_1, u_1, \dots, e_k, u_k$ to its vertex sequence $u_0, u_1, \dots, u_k$ is a one-to-one correspondence between the walks of length k and the sequences of $k+1$ vertices in which every two consecutive vertices are adjacent.*

Proof. This is proved here. The vertex sequence of a walk has consecutive vertices u_{i-1}, u_i joined by the edge e_i, so they are adjacent. The map is one to one: in a simple graph at most one edge joins two vertices, so $e_i = u_{i-1}u_i$ is determined by the vertices, and two walks with the same vertex sequence are equal. The map is onto: given such a sequence, put $e_i = u_{i-1}u_i$; this is an edge because the two vertices are adjacent, and the result is a walk. □

For $k = 0$ the same reasoning gives one walk from each vertex to itself, which is why the identity matrix $A^0 = I$ fits the pattern [3, Section 4.5.5]; this note uses it only in Proposition 4.17. Where simplicity enters the proof of Theorem 3.5 is stated after that proof.

C.2 The counterexample, in full

Proposition 3.10 uses the board graph. Its entries for the walks from n_1 to n_2 can be found in closed form. In B, the vertex n_2 is adjacent to n_1 and n_3, and these two are adjacent only to n_2. So a walk alternates between n_2 and the set $\{n_1, n_3\}$. A walk from n_1 to n_2 therefore has odd length k, and its vertices in even positions $w_2, w_4, \dots, w_{k-1}$, which number $(k-1)/2$, are each n_1 or n_3, freely. Hence there are $2^{(k-1)/2}$ walks of odd length k and none of even length, which gives $1, 0, 2, 0, 4$ for $k = 1, \dots, 5$, as in Table 2 and in the entries of A, A^3, A^5. Among them only the walk n_1, n_2 is a path or a trail, because any longer walk repeats the vertex n_2 (a walk of length 3 or more from n_1 to n_2 meets n_2 at an earlier odd position).

C.3 The chain of premises for the connectivity test

Theorem 4.6(a)⇔(b) follows from Proposition 4.5 and the definition of connected. The premises, in order, are the following.

1. Theorem 3.5: $(A^k)_{ij}$ is the number of walks of length k.
2. Corollary 3.9(a): all entries are nonnegative integers, so a sum is positive if and only if a summand is.
3. Lemma 2.44: a walk of length k from u to v contains a path of length at most k.
4. Corollary 2.46: a path in a graph with N vertices has at most $N-1$ edges.
5. Proposition 2.35 and Remark 2.49: "path" may be read as trail or path.

Premises 1 and 2 give (ii)⇔(iii) and the passage from a positive entry to a walk; premise 3 gives the passage from a walk to a path; premise 4 gives the bound $k \le N-1$ in (i)⇒(ii); premise 5 makes the equivalence independent of the reading of "path". No other fact is used.

C.4 Powers of $I + A$

Proof of Proposition 4.17. This is proved here; [1, Section 1, Theorem 2, proof sketch] gives the idea of adding a loop to every vertex. Put $m \ge N-1$. Expand $(I+A)^m = (I+A)(I+A)\cdots(I+A)$ with m factors. Each term chooses I or A in each factor and multiplies the choices in order. Since I times a matrix X is X, a term with k letters A equals A^k, and there are $\binom{m}{k}$ ways to choose which k factors give A. Therefore

$$(I + A)^m = \sum_{k=0}^{m} \binom{m}{k} A^k, \qquad A^0 = I. \tag{3}$$

Diagonal. The term $k = 0$ contributes 1 to every diagonal entry and all other terms are nonnegative, so every diagonal entry is at least 1. *Off the diagonal.* For $i \neq j$ the identity contributes 0, so $((I + A)^m)_{ij} = \sum_{k=1}^{m} \binom{m}{k} (A^k)_{ij}$. Every coefficient $\binom{m}{k}$ with $1 \leq k \leq m$ is a positive integer and every $(A^k)_{ij}$ is a nonnegative integer (Corollary 3.9(a)), so the sum is positive if and only if some $(A^k)_{ij}$ with $1 \leq k \leq m$ is positive. If a path between n_i and n_j exists, it has length k with $1 \leq k \leq N - 1 \leq m$ (Corollary 2.46) and $(A^k)_{ij} \geq 1$. Conversely, if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq m$, a walk exists (Theorem 3.5) and hence a path (Lemma 2.44). This proves the statement about the entries. G is connected if and only if every pair $i \neq j$ has a path, that is, if and only if no off-diagonal entry is 0; since the diagonal entries are positive, that is if and only if $(I + A)^m$ has no zero entry. Squaring $I + A$ repeatedly gives the exponents $2, 4, 8, \ldots$, so t squarings give $m = 2^t$, and t is chosen with $2^t \geq N - 1$. □

For $N = 3$ and the board graph, $(I + A)^2 = I + 2A + A^2$ is verified entry by entry in Example 6.2.

C.5 Distance

Proof of Proposition 2.47. This is proved here. Every path is a trail (Proposition 2.35) and every trail is a walk, so the lengths of paths are among the lengths of trails, which are among the lengths of walks; hence $\min_{\text{walks}} \leq \min_{\text{trails}} \leq \min_{\text{paths}}$. By Lemma 2.44 a shortest walk contains a path of at most the same length, so $\min_{\text{paths}} \leq \min_{\text{walks}}$. The three minima are therefore equal. If there is no walk there is no trail and no path, and if there is a walk, Lemma 2.44 gives a path; so the three sets are empty together. □

Proposition C.2 (the distance properties)**.** *The three properties listed in Definition 2.40 follow from the definition.*

Proof. The textbook lists the properties without proof [9, p. 23]; this proof is written here. By Proposition 2.47 we may compute d by shortest walks. (1) Lengths are $0, 1, 2, \ldots$ or $+\infty$, so $d \geq 0$. The walk consisting of u alone has length 0, so $d(u, u) = 0$. If $d(u, v) = 0$ there is a walk of length 0 from u to v, which is the single vertex, so $v = u$. (2) If $u_0, \ldots, u_k$ is a walk from u to v then $u_k, \ldots, u_0$ is a walk from v to u of the same length, since edges are unordered; so the sets of lengths agree, and so do their minima. (3) If $d(u, w)$ or $d(w, v)$ is $+\infty$ the right side is $+\infty$ and the inequality holds. Otherwise join a shortest (u, w)-walk and a shortest (w, v)-walk at w: this is a (u, v)-walk of length $d(u, w) + d(w, v)$, and a shortest (u, v)-walk is no longer. □

C.6 Cross-checks by computer

Two finite checks accompany the proofs; neither replaces them. First, for every labelled simple graph on $N = 2, 3, 4, 5$ vertices, the number of walks of length k enumerated by brute force equals the corresponding entry of A^k, for $k = 1, \ldots, 5$ when $N \leq 4$ and $k = 1, 2, 3$ when $N = 5$ (Appendix D). Second, for all labelled simple graphs on $2, \ldots, 6$ vertices, the sum test, the $(I + A)^{N-1}$ test, the component statement of Corollary 4.7, and the repeated-squaring form of Proposition 4.17 agree with breadth-first search (Table 6).

D Sources read and how to reproduce the numbers

Sources and what was read. *Textbook* [9]: printed pages 9 to 12, 16, 21 to 23, and 48 to 49, viewed as page images, and a full-text search of the whole book for the adjacency matrix; the other chapters were not read. *Board photographs* [4, 5]: both viewed as images. *Grinberg* [3]: Sections 2.9, 2.10, 3.4, and 4.5.5 of the 454-page text. *Galway notes* [2]: the whole eight-page file. *Lehman, Leighton, and Meyer* [6]: the one page cited. *Paramadevan and Sotheeswaran* [7]: the whole nine-page paper; it is an expository paper whose Theorem 1 and proofs are loose

(Section 7.4), so it is cited for statements and as a second witness only. *Bai* [1]: Section 1; the rest concerns eigenvalues. *MathWorld* [11] and *Wikipedia* [8]: the entries cited; the Wikipedia article is open to later edits. *Schost* [10]: the slides on breadth-first search and its running time. The assignment text is known only as it was reported to us and is not a source of the mathematics.

Scripts. All scripts are in the folder `appendices/scripts/`, are written in plain Python with integer arithmetic, and are run from the paper folder.

- `generate_paper_numbers.py` computes every matrix and table printed in the examples, asserts the facts the text states about them, and writes `numbers.tex`, which this note inputs.
- `m1_walks.py`: powers of the board matrix, explicit walk lists, the hand-checkable sums, walks against paths and trails, and a brute-force check of Theorem 3.5 on all labelled simple graphs with $N = 2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs; 40, 360, 5120, 76800 entries, all equal to the walk counts).
- `m2_walks.py`: classification of the example walks, the absence of a cycle in the board graph among closed walks of length up to 6, traces of Lemma 2.44, and checks of the lemma, Proposition 2.35, Corollary 2.46, and Proposition 2.47 on all walks of length up to 6 in three small graphs and on 300 random graphs (seed 20261027, 192696 walks, no failures). Its textbook example graph uses only seven edges that the textbook's examples use; the textbook's figure has at least one more, which no example uses.
- `m3_connectivity.py`: the examples of Section 6, the path graphs of Table 3, the cost counts, and the exhaustive comparison over all labelled graphs on up to 6 vertices, including the component statement of Corollary 4.7.

To reproduce, run `python3 -I appendices/scripts/generate_paper_numbers.py` and the three other scripts the same way; their saved outputs are in the same folder. The last script takes about twenty seconds.